

34th Conference On Neural Information Processing Systems (NeurIPS 2020)

Effective Diversity in Population Based Reinforce Learning

Jack Parker-Holder*

University of Oxford

Aldo Pacchiano*

UC Berkeley

Krzysztof Choromanski

Google Brain Robotics

2022/4/10

Presenter 이근배

Fundamental Team 김동현, 김채현, 양현모, 이재윤, 조남경



1. INTRODUCTION

2. DIVERSITY VIA DETERMINANTS

3. DvD ALGORITHM

4. EXPERIMENTS

1

1. Introduction

Markov Decision Process (MDP)

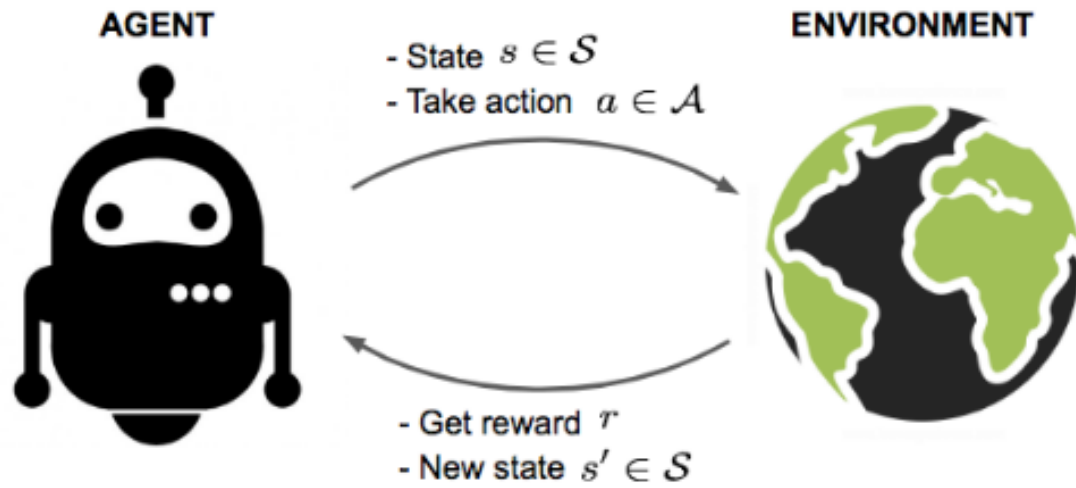


Fig. 1. An agent interacts with the environment, trying to take smart actions to maximize cumulative rewards.

On-policy RL

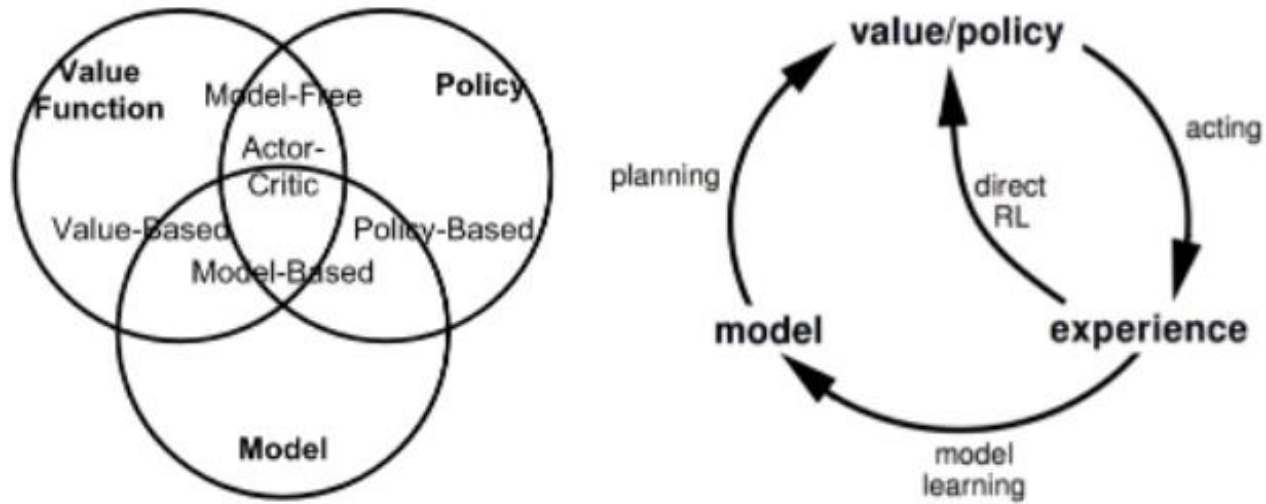


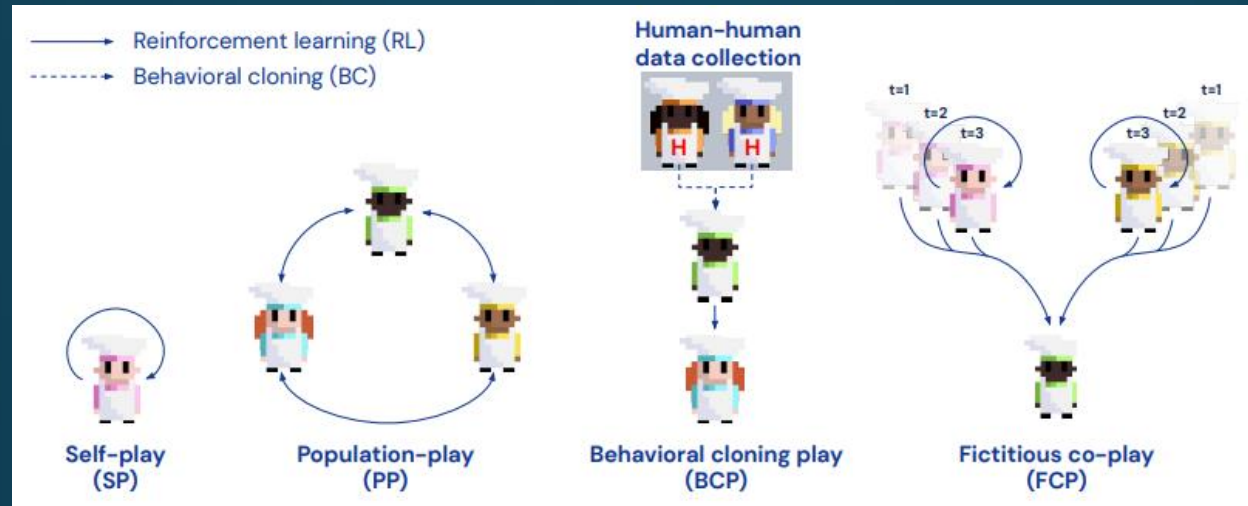
Fig. 2. Summary of approaches in RL based on whether we want to model the value, policy, or the environment. (Image source: reproduced from David Silver's RL course [lecture 1](#).)

$$J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau)]$$

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) R(\tau) \right]$$

$$J(\pi_\theta) = R(\tau_\theta) + \lambda d(\tau_\theta)$$

About Reinforced Learning



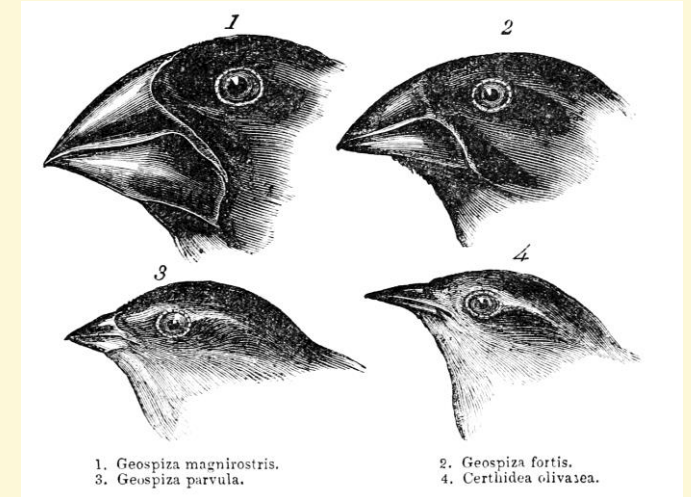
- Agent acts in an environment to maximize total reward
- Training a population of agents to gather a diverse range of experiences at the same training time
 - Population Based Training: Jointly optimizes NN weights & hyperparameters which periodically copies weights of the best performers, and updates hyperparameters during training³²
- Popular method in Neuroevolution → become popular in RL

Source: Figure 2. Strouse, D. J., et al. "Collaborating with humans without human data." *Advances in Neural Information Processing Systems* 34 (2021).

32: Li, Ang, et al. "A generalized framework for population based training." *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2019.

[Wikipedia]

- Use **evolutionary algorithms** to generate neural net, parameters, topology and rules
- Most commonly applied in artificial life, general game playing and evolutionary robotics
- Requires only a measure of a network's performance at a task
- I.e., the outcome of a game (win or lost) can be easily measured without providing labeled examples of desired strategies



Behavioral characterizations (BC)

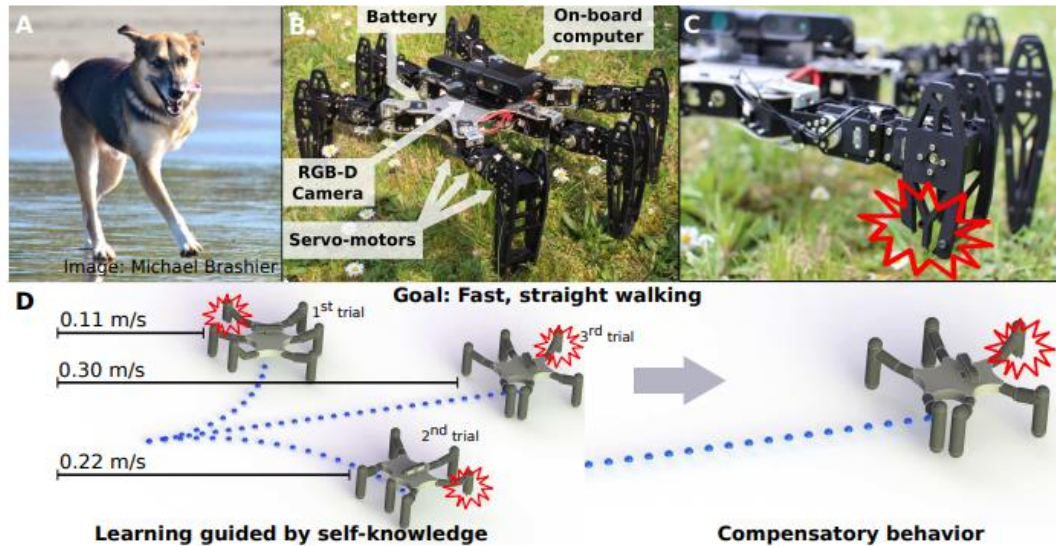


Figure 1 With Intelligent Trial and Error, robots, like animals, can quickly adapt to recover from damage. (A) Most animals can find a compensatory behavior after an injury. Without relying on *predefined* compensatory behaviors, they learn how to avoid behaviors that are painful or no longer effective. (B) An undamaged, hexapod robot. (C) One type of damage the hexapod may have to cope with (broken leg). (D) After damage occurs, in this case making the robot unable to walk straight, damage recovery via Intelligent Trial and Error begins. The robot tests different types of behaviors from an automatically generated map of the behavior-performance space. After each test, the robot updates its predictions of which behaviors will perform well despite the damage. This way, the robot rapidly discovers an effective compensatory behavior.

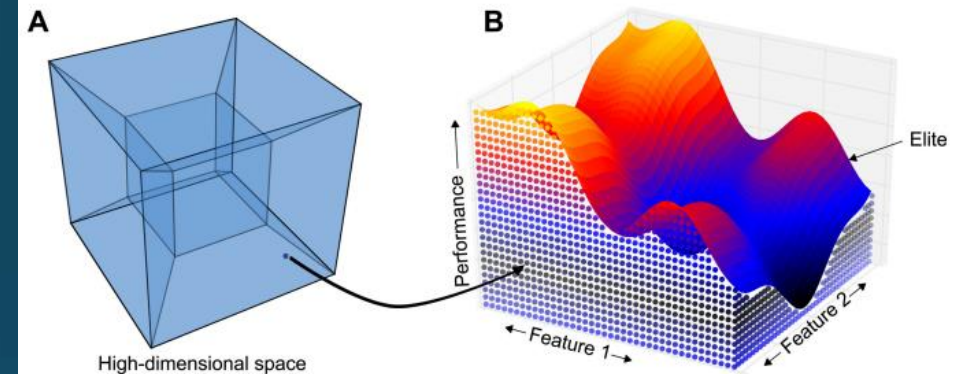


Fig. 1. The MAP-Elites algorithm searches in a high-dimensional space to find the highest-performing solution at each point in a low-dimensional feature space, where the user gets to choose dimensions of variation of interest that define the low dimensional space. We call this type of algorithm an “illumination algorithm”, because it illuminates the fitness potential of each area of the feature space, including tradeoffs between performance and the features of interest. For example, MAP-Elites could search in the space of all possible robot designs (a very high dimensional space) to find the fastest robot (a performance criterion) for each combination of height and weight.

Contribution

- Formalization of behavioral characterization (BC)
- Define behavioral embedding as the action taken by policies
- Measure the diversity of the entire population as the value of the similarity matrix → Introduce Diversity via Determinants (DvD)
- Implementation of DvD
 - Evolution strategies (DvD-ES) → able to discover diverse solutions in multi-modal environments
 - Off-policy RL (DvD-TD3) → proof that DvD still performs well when diversity is not required

Q&A



The background of the slide features three Erlenmeyer flasks arranged horizontally. A pipette is positioned above the leftmost flask, with a single drop of liquid falling into it. The flasks contain a light blue liquid and a dark, clumpy substance at the bottom. The entire scene is set against a solid blue background.

2. Diversity via Determinants

Task Agnostic Behavioral Embeddings

Definition 3.1. Let θ^i be a vector of neural network parameters encoding a policy π_{θ^i} and let $\mathcal{S}$ be a finite set of states. The **Behavioral Embedding** of θ^i is defined as: $\phi(\theta^i) = \{\pi_{\theta^i}(\cdot|s)\}_{s \in \mathcal{S}}$.

This approach allows us to represent the behavior of policies in vectorized form, as $\phi : \theta \rightarrow \mathbb{R}^l$, where $l = |a| \times N$, where $|a|$ is the dimensionality of each action $a \in \mathcal{A}$ and N is the total number of states. When N is the number of states in a finite MDP, the policies are deterministic and the embedding is the as in Definition 3.1, we have:

$$\phi(\theta^i) = \phi(\theta^j) \iff \pi_{\theta^i} = \pi_{\theta^j}, \quad (4)$$

where θ^i, θ^j are vectorized parameters. In other words, the policies are the same since they always take the same action in every state of the finite MDP. Note that this does not imply that $\theta^i = \theta^j$.

Joint Population Update

$$k_{SE}(x_1, x_2) = \exp \left(-\frac{\|x_1 - x_2\|^2}{2l^2} \right),$$

$$k(\phi(\theta^i), \phi(\theta^j)) = 1 \iff \pi_{\theta^i} = \pi_{\theta^j}$$

Definition 3.2. (Population Diversity) Consider a finite set of M policies, parameterized by $\Theta = \{\theta^1, \dots, \theta^M\}$, with $\theta^i \in \mathbb{R}^d$. We denote $\text{Div}(\Theta) \stackrel{\text{def}}{=} \det(K(\phi(\theta_t^i), \phi(\theta_t^j))_{i,j=1}^M) = \det(\mathbf{K})$, where $K : \mathbb{R}^l \times \mathbb{R}^l \rightarrow \mathbb{R}$ is a given kernel function. Matrix $\mathbf{K}$ is positive semidefinite since all principal minors of $\det(\mathbf{K})$ are nonnegative.

Now consider the DvD loss function, as follows:

$$J(\Theta_t) = \underbrace{\sum_{i=1}^M \mathbb{E}_{\tau \sim \pi_{\theta^i}} [R(\tau)]}_{\text{individual rewards}} + \underbrace{\lambda_t \text{Div}(\Theta_t)}_{\text{population diversity}}$$

$$J(\pi_\theta) = R(\tau_\theta) + \lambda d(\tau_\theta)$$

Theorem 3.3. Let M be a finite, tabular MDP with $\tilde{M} \geq M$ distinct optimal policies $\{\pi_i\}_{i=1}^{\tilde{M}}$ all achieving a cumulative reward of $\mathcal{R}$ and such that the reward value $\mathcal{R}(\pi)$ of any suboptimal policy π satisfies $\mathcal{R}(\pi) + \Delta < \mathcal{R}$ for some $\Delta > 0$. There exists $\lambda_t > 0$, such that the objective in Equation [7](#) can only be maximized if the population contains M distinct optimal solutions.

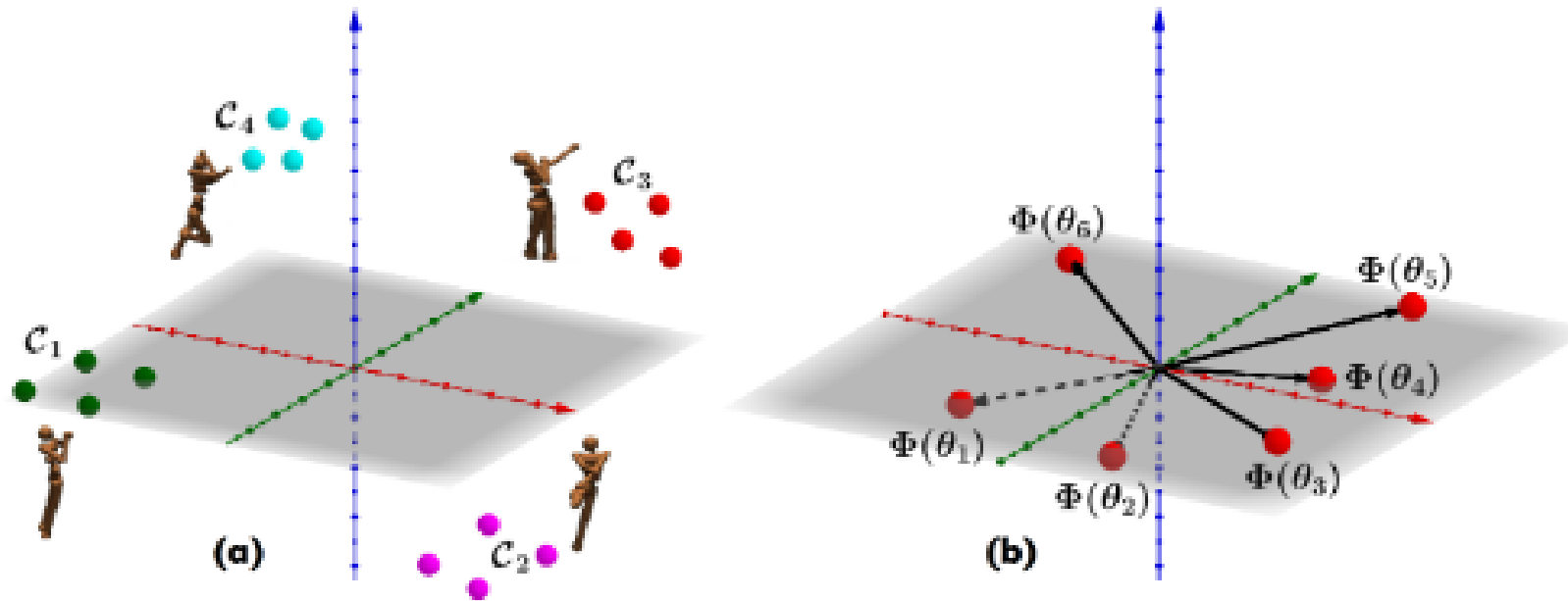
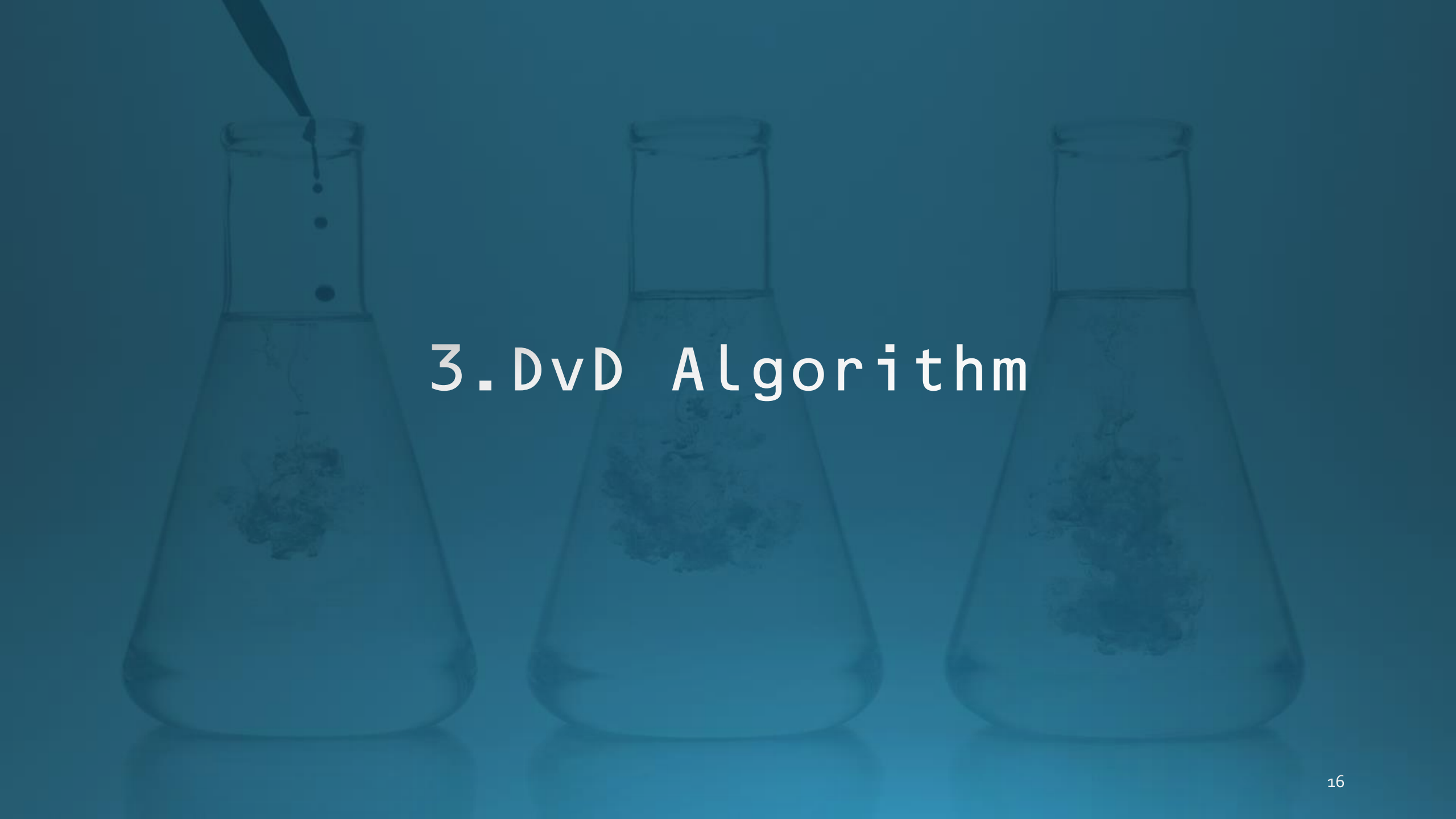


Figure 1: Determinant vs. pairwise distance. (a): populations of agents split into four clusters with agents within cluster discovering similar policies. (b): embedded policies $\phi(\theta_1), \dots, \phi(\theta_6)$ lie in a grey hyperplane. In (a) resources within a cluster are wasted since agents discover very similar policies. In (b) all six embeddings can be described as linear combinations of embeddings of fewer canonical policies. In both settings the mean pairwise distance will be high but diversity as measured by determinants is low.

Q&A



The background of the slide features three Erlenmeyer flasks arranged horizontally. The flasks are filled with a light blue liquid. A pipette is positioned above the leftmost flask, with a single drop of liquid falling into it. The entire scene is set against a blue gradient background that transitions from a darker blue at the top to a lighter blue at the bottom. The text '3. DvD Algorithm' is centered over the middle flask.

3. DvD Algorithm

Approximate Embeddings & Adaptive Exploration

In most practical settings the state space is intractably or infinitely large. Therefore, we must sample the states $\{s_i\}_{i=1}^n$, where $n < N$, and compute the embedding as an expectation as follows:

$$\hat{\phi}(\theta_i) = \mathbb{E}_{s \sim \mathcal{S}}[\{\pi_{\theta^i}(\cdot|s)\}] \quad (8)$$

Now consider the DvD loss function, as follows:

$$J(\Theta_t) = \underbrace{\sum_{i=1}^M \mathbb{E}_{\tau \sim \pi_{\theta^i}}[R(\tau)]}_{\text{individual rewards}} + \underbrace{\lambda_t \text{Div}(\Theta_t)}_{\text{population diversity}}$$

DvD-ES & DvD-TD3 Algorithm

$$\Theta_t = \{\theta_t^i\}_{i=1}^M \quad \{\mathbf{g}_i^m\}_{i=1, \dots, k}^{m=1, \dots, M}$$

DvD-ES

$$\text{Div}_t(i) = \text{Div}_t(\theta_t^1 + \mathbf{g}_i^1, \dots, \theta_t^M + \mathbf{g}_i^M). \quad (9)$$

Thus, the DvD-ES gradient update is the following:

$$\theta_{t+1}^m = \theta_t^m + \frac{\eta}{k\sigma} \sum_{i=1}^k [(1 - \lambda_t) R_i^m + \lambda_t \text{Div}_t(i)] \mathbf{g}_i^m. \quad (10)$$

DvD-TD3

Now consider the DvD loss function, as follows:

$$J(\Theta_t) = \underbrace{\sum_{i=1}^M \mathbb{E}_{\tau \sim \pi_{\theta^i}} [R(\tau)]}_{\text{individual rewards}} + \underbrace{\lambda_t \text{Div}(\Theta_t)}_{\text{population diversity}}$$

Lemma 4.1. *The gradient of $\log(\det(\mathbf{K}))$ with respect to $\Theta = \theta^1, \dots, \theta^M$ equals: $\nabla_{\theta} \log(\det(\mathbf{K})) = -(\nabla_{\theta} \Phi(\theta)) (\nabla_{\Phi} \mathbf{K}) \mathbf{K}^{-1}$, where $\phi(\theta) = \phi(\theta^1) \dots \phi(\theta^M)$.*

Q&A



The background of the slide features three Erlenmeyer flasks arranged horizontally. A pipette is positioned above the leftmost flask, with a single drop of liquid falling into it. The flasks contain a light blue liquid and a dark, clumpy substance at the bottom. The entire scene is set against a solid blue background.

4. Experiments

Finding Diverse Solutions with DvD-ES

Exploration

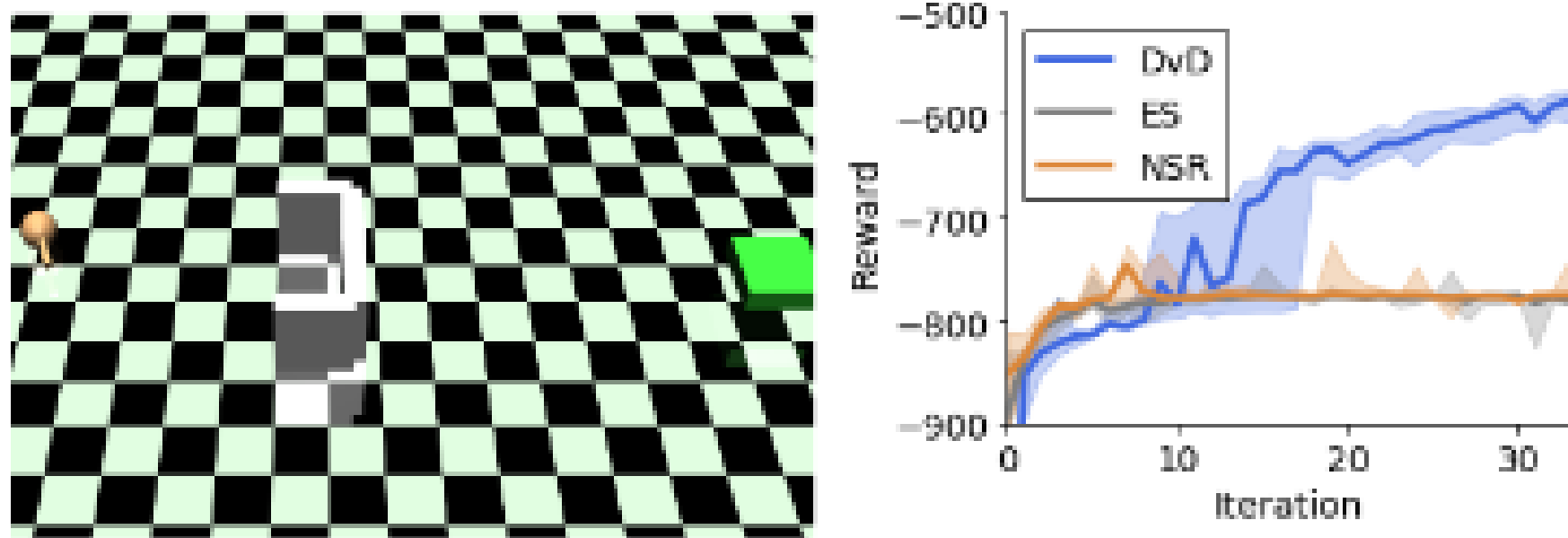


Figure 3: Left: the Point-v0 environment. Right: median best performing agent across ten seeds. IQR shaded.

Backup: MuJoCo <https://gym.openai.com/envs/#mujoco>

MuJoCo

Continuous control tasks, running in a fast physics simulator.



Ant-v2
Make a 3D four-legged robot walk.



HalfCheetah-v2
Make a 2D cheetah robot run.



Hopper-v2
Make a 2D robot hop.



Humanoid-v2
Make a 3D two-legged robot walk.



HumanoidStandup-v2
Make a 3D two-legged robot standup.



InvertedDoublePendulum-v2
Balance a pole on a pole on a cart.



InvertedPendulum-v2
Balance a pole on a cart.



Reacher-v2
Make a 2D robot reach to a randomly located target.



Swimmer-v2
Make a 2D robot swim.



Walker2d-v2
Make a 2D robot walk.

Finding Diverse Solutions with DvD-ES

Multi-Modal Environments

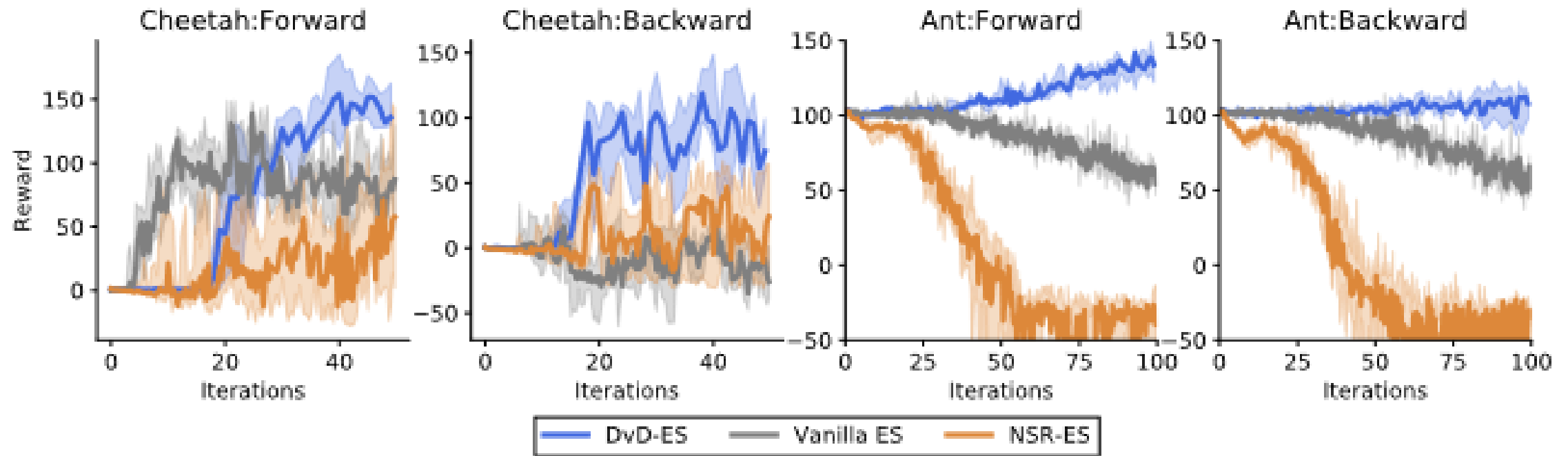


Figure 4: The median best performing agent across ten seeds for multi-modal tasks and two environments: Cheetah and Ant. The plots show median curves with IQR shaded.

Finding Diverse Solutions with DvD-ES

Single Mode Environments

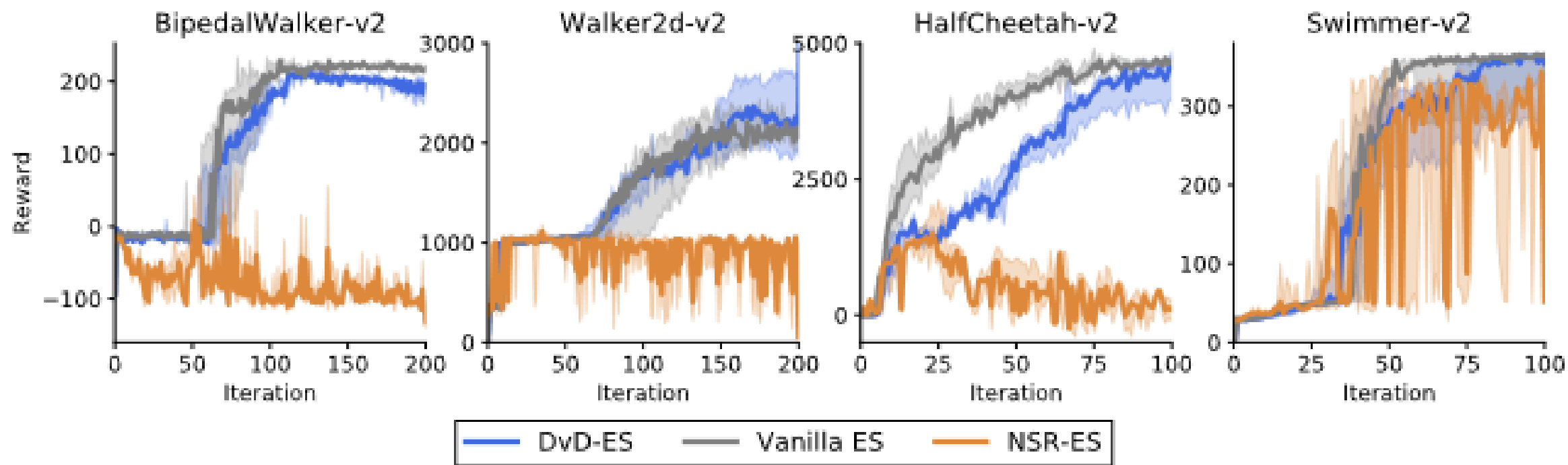


Figure 5: The median best performing agent across five seeds for four different environments.

Choice of Kernel

Table 1: This table shows the median maximum performing agent across 5 seeds. All algorithms shown are identical aside from the choice of DPP kernel. Training is for $\{50, 100, 200\}$ iterations for $\{\text{point}, \text{SwimmerWalker2d}\}$ respectively.

	Point	Swimmer	Walker2d
Squared Exponential	-547.03	354.86	1925.86
Exponential	-561.13	362.83	1929.81
Linear	-551.48	354.37	1944.95
Rational Quadratic	-548.55	246.68	2113.02
Matern $\frac{5}{2}$	-578.05	349.52	1981.66
Matern $\frac{3}{2}$	-557.69	357.88	1866.56

Finding Diverse Solutions with DvD-ES

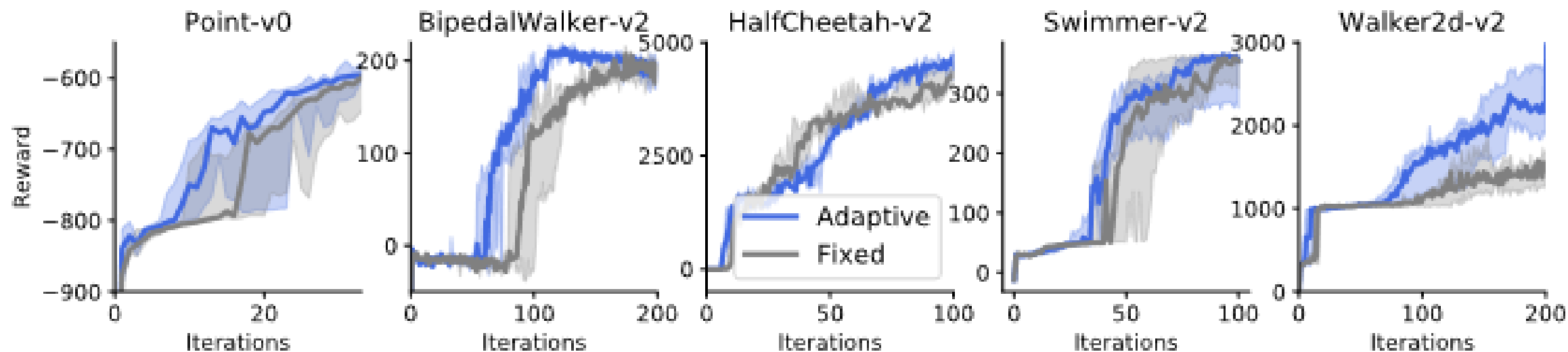
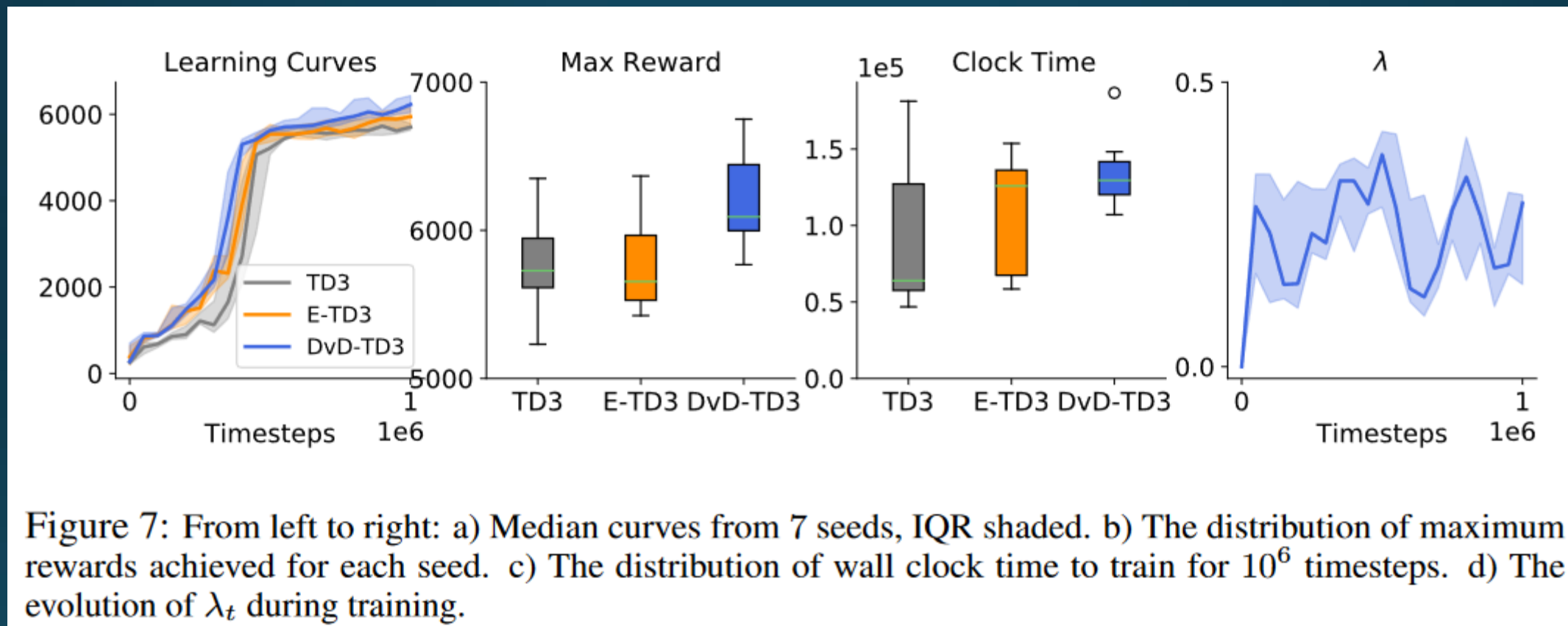


Figure 6: The median best performing agent across five seeds for five different environments.

Teaching a Humanoid to Run with DvD-TD3



Key Takeaways

Behavioral embedding

Population based training

Diversity in multi modal environments

Q&A

